

E.G.S.PILLAY ENGINEERING COLLEGE,(Autonomous)

Approved by AICTE, New Delhi Affiliated to Anna University, Chennai| Accredited by NAAC with,,
A++Grade Accredited by NBA(Tier-1)
NAGAPATTINAM–611002



B.E-Biomedical Engineering R-2023

CURRICULUM

Total Credits: 165 (Lateral Entry: 121)

S.NO	CATEGORY	CREDITS
A	Humanity & Social science including management course(HSMC)	5
	Basic Science (BSC)	24
	Engineering Science (ESC)	22
B	Professional Core Courses(PCC)	66
C	Professional Elective Courses(PE)	18
	Open Electives(OE)	9
D	Employment Enhancement Courses(Project work, Mini project, internship presentation, PDC, Industrial certification)	21
E	Mandatory Courses	Non Credit (Not included for CGPA)
	Total Credits to be earned for the award of the Degree	165 (from A to E) and the successful completion of Mandatory Courses

FIRST SEMESTER									
COURSE CODE	COURSENAME	CATEG ORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
Theory Courses									
2301IP101	Induction Program	-	0	0	0	0	0	0	0
2301MA102	Linear Algebra, Calculus and Partial Differentiation	BSC	3	1	0	4	40	60	100
2301PH101	Physics of Semiconductor and Optoelectronics Devices	BSC	3	0	0	3	50	50	100
2301CH101	Chemistryfor BiomedicalEngineering	BSC	3	0	0	3	50	50	100
2301GEX04	Problem Solving using Python	ESC	2	0	4	4	50	50	100
2301ENX01	Professional English	HSMC	2	0	0	2	50	50	100
2301TA101	Heritage of Tamils / தமிழர் மரபு	HSMC	1	0	0	1	100	0	100
Practical Courses									
2301PHX51	Engineering Physics Laboratory	BSC	0	0	2	1	60	40	100
2301CHX51	Engineering Chemistry Laboratory	BSC	0	0	2	1	60	40	100
2301ENX51	Communication Skills Laboratory	HSMC	0	0	2	1	100	0	100
2301GEX52	Engineering Practices Laboratory	ESC	0	0	4	2	60	40	100
2301LS101	Life Skills – I	-	0	0	0	0	100	0	100
TOTAL			14	1	14	22	720	380	1100
SECOND SEMESTER									
Theory Courses									
2301MA202	Calculus, Ordinary Differential Equations and Complex Variables	BSC	3	1	0	4	40	60	100
2302EC201	Electron Devices	ESC	3	0	0	3	40	60	100
2301GEX05	Applied Digital Logic and Design	ESC	3	0	0	3	40	60	100
2301GEX02	Engineering Graphics	ESC	2	1	0	3	40	60	100
2301GEX03	Problem Solving using C	ESC	2	0	4	4	50	50	100
2304FLX02	Language Elective - Japanese	EEC	2	0	0	2	100	0	100
2301TA201	Tamils and Technology/ தமிழரும் தொழில் துட்பமும்	HSMC	1	0	0	1	100	0	100
Practical Courses									
2301GEX51	Computer Practices Laboratory	ESC	0	0	2	1	100	0	100
2302BM251	Electron Devices Laboratory	ESC	0	0	2	1	60	40	100
2301GEX54	Applied Digital Logic and Design Laboratory	ESC	0	0	2	1	100	0	100
2301LS201	Life Skills–II	-	0	0	0	0	100	0	100
TOTAL			16	2	10	23	770	330	1100
THIRD SEMESTER									
Theory Courses									
2301MA302	Probability and Stochastic Processes	BSC	3	1	0	4	40	60	100
2302BM301	Biosciences in Medical Engineering	PCC	3	0	2	4	50	50	100
2302BM302	Bio Process Control Systems	PCC	3	0	0	3	40	60	100
2302BM303	Biomedical Circuits and Networks	PCC	3	0	2	3	40	60	100
2302BM304	Human Anatomy and Physiology	PCC	3	0	2	3	40	60	100
2301GEX07	Environmental Science and Sustainability	BSC	1	0	2	2	40	60	100
Practical Courses									
2302BM351	Human Anatomy and Physiology	PCC	0	0	2	1	60	40	100

	Laboratory								
2302BM352	Circuits and networks Laboratory	PCC	0	0	2	1	60	40	100
2304GE301	Professional Development Course-I	EEC	0	0	2	1	100	--	100
TOTAL			16	1	14	22	470	430	900
FOURTH SEMESTER									
Theory Courses									
2302BM401	Bio Sensors and Measurements	PCC	3	0	0	3	40	60	100
2302BM402	Communication Engineering for Biomedical Engineers	PCC	3	0	0	3	40	60	100
2302BM403	Biomedical Signals & Systems	PCC	3	1	0	4	40	60	100
2302BM404	Diagnostic and Therapeutic Equipment I	PCC	3	0	2	3	40	60	100
2302BM405	Biomedical Instrumentation	PCC	3	0	2	3	40	60	100
2301HSX01	Universal Human values and Ethics	BSC	1	0	2	2	40	60	100
2301MC402	Mandatory Course -1(Circular Economy)	MC	3	0	0	0	100	0	100
Practical Courses									
2302BM451	Biosensors and Measurements Laboratory	PCC	0	0	2	1	40	60	100
2302BM452	Diagnostic and Therapeutic Equipment Laboratory	PCC	0	0	2	1	40	60	100
2302BM453	Biomedical Instrumentation Laboratory	PCC	0	0	2	1	40	60	100
2304GE401	Professional Development Course - II	EEC	0	0	2	1	100	--	100
TOTAL			19	1	14	22	560	540	1100
FIFTH SEMESTER									
Theory Courses									
2302BM501	Diagnostic and Therapeutic Equipment II	PCC	3	0	0	3	40	60	100
2302BM502	Fundamental of Rehabilitation Healthcare Analytics	PCC	3	0	0	3	40	60	100
2302BM503	Bio Signal Processing	PCC	3	1	0	4	40	60	100
	Professional Elective I	PEC	3	0	0	3	40	60	100
	Professional Elective II	PEC	3	0	0	3	40	60	100
	Open Elective I	OEC	3	0	0	3	40	60	100
Practical Courses									
2302BM551	Biomechanics Laboratory	PCC	0	0	2	1	60	40	100
2302BM552	Biomedical Digital Signal Processing Laboratory	PCC	0	0	2	1	60	40	100
2304GE501	Professional Development Course-III	EEC	0	0	2	1	100	--	100
TOTAL			18	1	06	22	460	440	900
SIXTH SEMESTER									
Theory Courses									
2302BM601	Digital Image Processing	PCC	3	0	0	3	40	60	100
2302BM602	Bio Embedded Systems	PCC	3	0	0	3	40	60	100
2302BM603	Biomedical IOT	PCC	3	0	0	3	40	60	100
	Professional Elective III	PEC	3	0	0	3	40	60	100
	Professional Elective IV	PEC	3	0	0	3	40	60	100
	Open Elective II	OEC	3	0	0	3	40	60	100
2301MC605	Mandatory course-II (Well -being with traditional practices – Yoga, Ayurveda and Siddha)	MC	3	0	0	0	100	0	100
Practical Courses									
2302BM651	Image Processing Laboratory	PCC	0	0	2	1	40	60	100
2302BM652	Bio Embedded laboratory	PCC	0	0	2	1	40	60	100

2302BM653	Mini Project	EEC	0	0	2	1	40	60	100
2304GE601	Professional Development Course-IV	EEC	0	0	2	1	100	--	100
TOTAL			18	00	8	22	460	540	1100
SEVENTH SEMESTER									
Theory Courses									
2302BM701	AI/ML for Bio Medical Engineers	PCC	3	0	0	3	40	60	100
2302BM702	Medical Ethics and standards	PCC	3	0	0	3	40	60	100
2302BM703	Hospital Automation	PCC	3	0	0	3	40	60	100
	Professional Elective V	PCC	3	0	0	3	40	60	100
	Professional Elective VI	PEC	3	0	0	3	40	60	100
	Open Elective III	OEC	3	0	0	3	40	60	100
Practical Courses									
2302BM751	Project Work (Phase I)	EEC	0	0		4	50	50	100
2302BM752	Industrial certification	EEC	0	0	0	1	60	40	100
TOTAL			15	00		23	440	460	900
EIGHTH SEMESTER									
Practical Courses									
2302BM851	Internship Presentation	EEC	0	0	0	1	100	-	100
2302BM852	Project Work (Phase II)	EEC	0	0		8	50	50	100
TOTAL			00	00		09	150	50	200

BOS CHAIRMAN

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
PROFESSIONAL ELECTIVE – I									
2303BM001	Biomaterials	PEC	3	0	0	3	40	60	100
2303BM002	Foundation Skills in Integrated Product Development	PEC	3	0	0	3	40	60	100
2303BM003	Clinical Engineering	PEC	3	0	0	3	40	60	100
2303BM004	Biomechanics	PEC	3	0	0	3	40	60	100
2303BM005	Speech and Audio Signal Processing	PEC	3	0	0	3	40	60	100
2303BM006	Medical Optics	PEC	3	0	0	3	40	60	100
PROFESSIONAL ELECTIVE – II									
2303BM007	Artificial Organs and Implants	PEC	3	0	0	3	40	60	100
2303BM008	Medical Device Design	PEC	3	0	0	3	40	60	100
2303BM009	Hospital Planning and Management	PEC	3	0	0	3	40	60	100
2303BM010	Rehabilitation Engineering	PEC	3	0	0	3	40	60	100
2303BM011	Biometrics	PEC	3	0	0	3	40	60	100
2303BM012	Wearable Devices	PEC	3	0	0	3	40	60	100
PROFESSIONAL ELECTIVE - III									
2303BM014	Radiological Equipment's	PEC	3	0	0	3	40	60	100
2303BM015	Patient safety, Standards and Ethics	PEC	3	0	0	3	40	60	100
2303BM016	Medical Waste Management	PEC	3	0	0	3	40	60	100
2303BM017	Physiological Modeling	PEC	3	0	0	3	40	60	100
2303BM018	Pattern recognition techniques	PEC	3	0	0	3	40	60	100
2303BM019	Body Area Networks	PEC	3	0	0	3	40	60	100
PROFESSIONAL ELECTIVE - IV									
2303BM020	Neural Engineering	PEC	3	0	0	3	40	60	100
2303BM021	Medical Device Regulations	PEC	3	0	0	3	40	60	100
2303BM022	Economics and Management for Engineers	PEC	3	0	0	3	40	60	100
2303BM023	Human Assistive Devices	PEC	3	0	0	3	40	60	100
2303BM024	Brain Computer Interface and Applications	PEC	3	0	0	3	40	60	100
2303BM025	Virtual Reality and Augmented Reality in Healthcare	PEC	3	0	0	3	40	60	100
PROFESSIONAL ELECTIVE - V									
2303BM026	Principles of Tissue Engineering	PEC	3	0	0	3	40	60	100
2303BM027	Medical Innovation and Entrepreneurship	PEC	3	0	0	3	40	60	100
2303BM028	Bio statistics	PEC	3	0	0	3	40	60	100
2303BM029	Ergonomics	PEC	3	0	0	3	40	60	100
2303BM030	Medical Imaging Systems	PEC	3	0	0	3	40	60	100
2303BM031	Tele health Technology	PEC	3	0	0	3	40	60	100
PROFESSIONAL ELECTIVE - VI									
2303BM032	Genetic Engineering	PEC	3	0	0	3	40	60	100
2303BM033	Rapid Prototyping	PEC	3	0	0	3	40	60	100
2303BM034	Forensic Science in Healthcare	PEC	3	0	0	3	40	60	100
2303BM035	Haptics	PEC	3	0	0	3	40	60	100
2303BM036	Computer Vision	PEC	3	0	0	3	40	60	100
2303BM037	Medical Informatics	PEC	3	0	0	3	40	60	100
OPEN ELECTIVE									
2303BM013	Occupational safety and health in public health emergencies	OE	3	0	0	3	40	60	100
2303BM038	Hospital management	OE	3	0	0	3	40	60	100
2303BM039	Ambulance and emergency medical service management	OE	3	0	0	3	40	60	100